

Elly Huang

.NET Backend Engineer · AI-assisted Product Builder · Open to 100% Remote

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SUMMARY

.NET backend engineer with production experience in ASP.NET Core Web API, EF Core, SQL Server, LINE Messaging API, CI/CD, and AI-assisted development. Independently built and maintained backend systems serving **24,000+ LINE users**, including member binding, OTP, points, coupon redemption, Flex Message / Rich Menu flows, and multi-database business logic. Also shipped a cross-platform Flutter + Firebase app to the App Store and web. Cross-domain background in plant-science research, HR, and hospitality helps me work across ambiguous requirements, technical implementation, and real user workflows.

EXPERIENCE

Leading Engineer · .NET Backend — Taiwan Mobile

2025/12 – Present · Taipei

- Refactored a legacy .NET Core 3.1 public-facing URL-shortening service into a **.NET 8 ASP.NET Core Web API** — replacing SFTP/FTP batch jobs with REST endpoints, externalizing configuration for containerized deployment, and fixing three pre-existing bugs — and produced the initial Docker / Kubernetes packaging for the migration.
- Co-designed a **multi-zone ingress architecture** (Internet load balancer → DMZ Nginx reverse proxy → intranet Kubernetes) for on-prem migration; documented design decisions, trade-offs, and risks using spec-driven development, and validated the reverse-proxy design with local Podman PoCs.
- Coordinated infrastructure provisioning end-to-end across network, security, procurement, and platform teams — L4, DNS, TLS certificates, container registry, GitOps, firewall, and log platform — and gained **hands-on exposure to Kubernetes deployment and troubleshooting** in a restricted enterprise environment (admission policies, cross-zone DNS / firewall connectivity).
- Analyzed a legacy .NET Framework telecom operations system to move risk-alerting logic to database-driven calculation: mapped 30+ report fields to schema and **validated assumptions against production data with SQL, disproving three initial assumptions**, then prepared structured cross-team requirement-alignment documentation.

.NET 8 / .NET Framework · ASP.NET Core Web API · Kubernetes · Docker / Podman · Nginx · Redis · SQL

.NET Backend Engineer — Universal Vision Biotechnology

2024/7 – 2025/11 · Taipei

- Independently built and maintained a **LINE Official Account backend serving 24,000+ users** — member binding, OTP, points, stamp collection and coupon redemption.
- Implemented **35 API endpoints across 3 databases** with ASP.NET Core 8, EF Core, LINQ, stored procedures, and cross-database business logic.
- Built secure webhook / API flows: LINE HMAC-SHA256 signature verification, Swagger IP filtering, duplicate-request protection, SMS OTP rate limiting, and binding-history audit trails.
- Designed modular **Flex Message** templates, dynamic **Rich Menu** switching by member state, and personalized LINE notifications for business engagement flows.
- Delivered an essay-contest management backend: **13 core APIs, 11 workflow states, 3 submission plans**, dual-layer review, URL deduplication by submission status, JWT / RBAC-protected endpoints, and secure file-upload validation.
- Built **CI/CD with automated deploy and rollback** using GitHub Actions; collaborated with UI/UX, planning, and frontend teams to turn business workflows into API contracts.

.NET Core · EF Core · LINQ · MS SQL Server · LINE Messaging API · LIFF

Full-Stack Engineer — Data Express

2024/2 – 2024/6 · Taipei

- Maintained and developed internal systems: a **BPM e-form system** (approval workflows incl. travel-expense forms), an EIP system, and the company marketing website (.NET MVC).
- Backend work on the corporate site (.NET WebAPI); MS SQL CRUD operations.

Research Assistant — Academia Sinica

2020/3 – 2023/7 · Taipei

- Led two large research projects spanning four collaborating labs; defined experimental workflows for unfamiliar plant species / techniques and applied hypothesis-driven analysis.
- Built strong habits in literature review, ambiguity reduction, technical documentation, and cross-lab coordination without formal authority.

SELECTED BACKEND / PRODUCT PROJECTS

LINE Bot Member Engagement Backend

Universal Vision Biotechnology

- Production LINE Official Account backend for 24,000+ users, integrating member identity, OTP verification, account binding, points, stamp collection, coupon lifecycle, Webhook event logging, and personalized Flex Message notifications.
- Architecture: layered ASP.NET Core Web API, dependency injection, EF Core Database-First, SQL Server, third-party SMS API, LINE Messaging API / LIFF, structured logging, and GitHub Actions deployment workflows.

Essay Contest Management System

Universal Vision Biotechnology

- Enterprise workflow backend for social-media / review submissions: 13 core APIs, 11 status states, 3 plan types, dual LINE Bot architecture, consultant review, customer-service final review, status history, and notification events.
- Implemented status-aware URL duplicate checks, plan-specific submission validation, file upload restrictions, JWT authentication, role-based access control, and scoped query logic so consultants only see assigned customer submissions.

TaskTree Cross-Platform App

Solo project · App Store + Web

- Built and shipped a Flutter + Firebase task-management app with 6-level hierarchical tasks, a three-state Active / Completed / Paused state machine, recursive cascade rules, anonymous-to-Google account upgrade, bilingual i18n, and Firebase Hosting deployment.

EARLIER EXPERIENCE

- **Project Manager**, Eurasia Tea (2019/10–2020/2) — inventory / operations at a tea-trade startup; supported early e-commerce listings.
- **HR Specialist**, Caves Books (2016/12–2018/3) — full-cycle recruiting; processed **7,000+ résumés**; onboarding and internal e-learning.
- **Front Desk Assistant Manager**, Island Paradise Resort, Palau (2015/7–2016/6) — managed a **20-person multinational team**; created soft-opening SOPs that improved efficiency by **80%**, with <1% error rate.

SKILLS

Core: .NET 8 / .NET Core · C# · Entity Framework Core · ASP.NET Core Web API · MS SQL Server / PostgreSQL · Redis · CI/CD (GitHub Actions) · LINE Messaging API · Flutter · AI-assisted Development · Cross-functional Collaboration

Exposure: Docker / Podman · Kubernetes · Nginx reverse proxy · ArgoCD GitOps · React

Recent focus: backend modernization with growing hands-on exposure to containerization and Kubernetes deployment.

EDUCATION

- **MSc, Plant Science** — National Taiwan University
- **BSc, Horticulture & Landscape Architecture** — National Taiwan University

LANGUAGES

Mandarin (native) · English (TOEIC 860) · a little Polish

SELECTED WORK

- **Portfolio** — aibuilderfolio.web.app
- **TaskTree** — cross-platform task app (App Store + web), built solo with Flutter + Firebase — growmytasktree.web.app